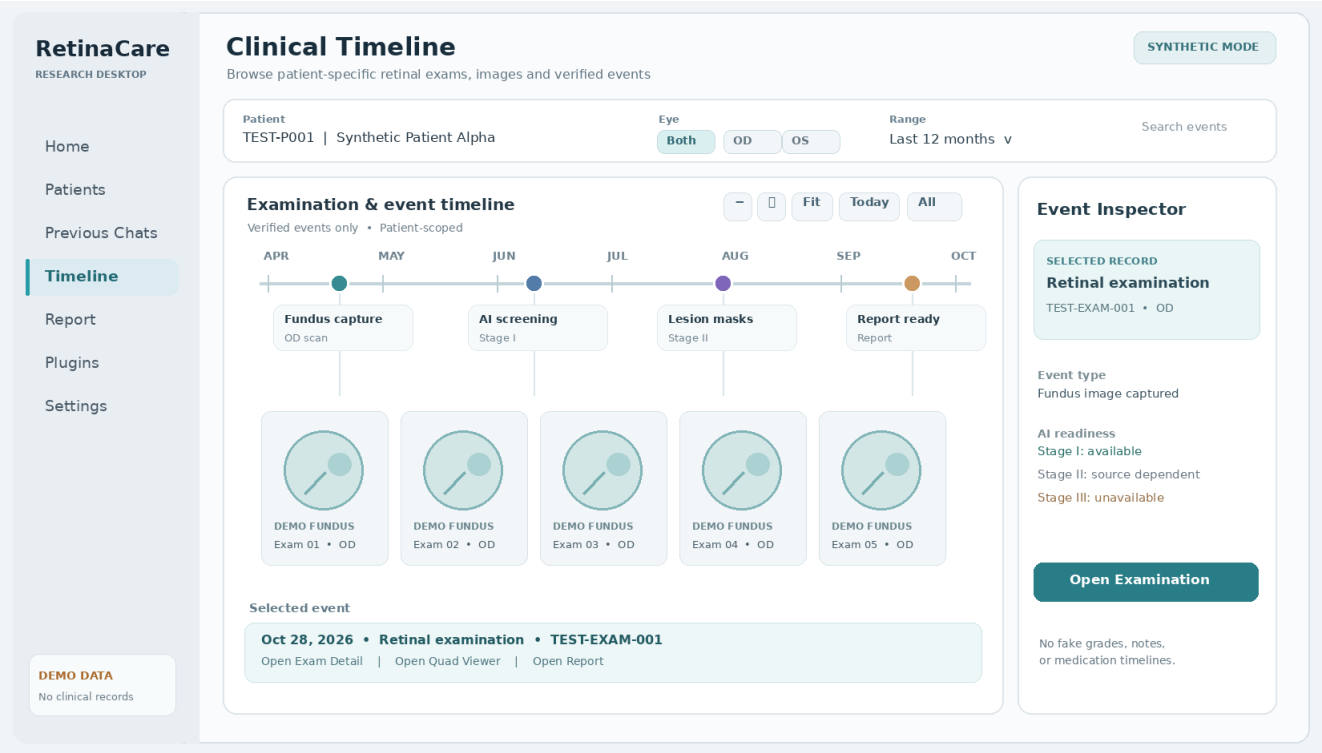
Clinical Timeline - Dedicated Sidebar Page

A design and implementation ADDENDUM to Task 07 Previous Chats & History. New retinal examination timeline, influenced by your uploaded screenshot, but adapted for a softer, readable native Rust interface.

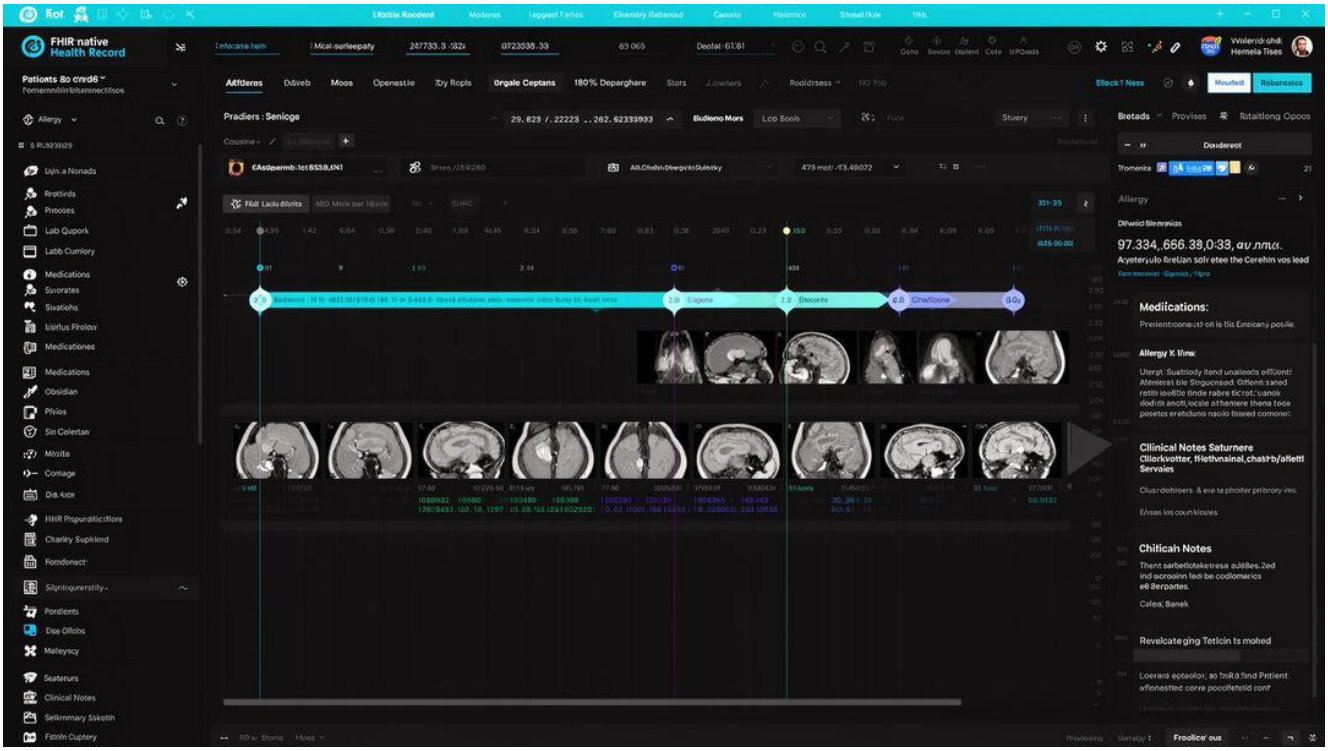


Concept design only. All shown patients, dates, examinations, thumbnails and event statuses are clearly synthetic placeholders; the running GUI must use authorized data or an honest unavailable state.

OWNER	DELIVERABLE	INTEGRATION RULE
Antigravity	New Timeline sidebar page + functional history UI	Do not edit Claude backend
Existing Task 07	Previous Chats & History remains separate	Extend; do not replace
Claude backend	Provides real events and persistence later	No invented clinical records

Priority: Plan first, then implement the new real native page; validate actual window sizes and clickable controls, update GUI Obsidian notes, report honest backend gaps.

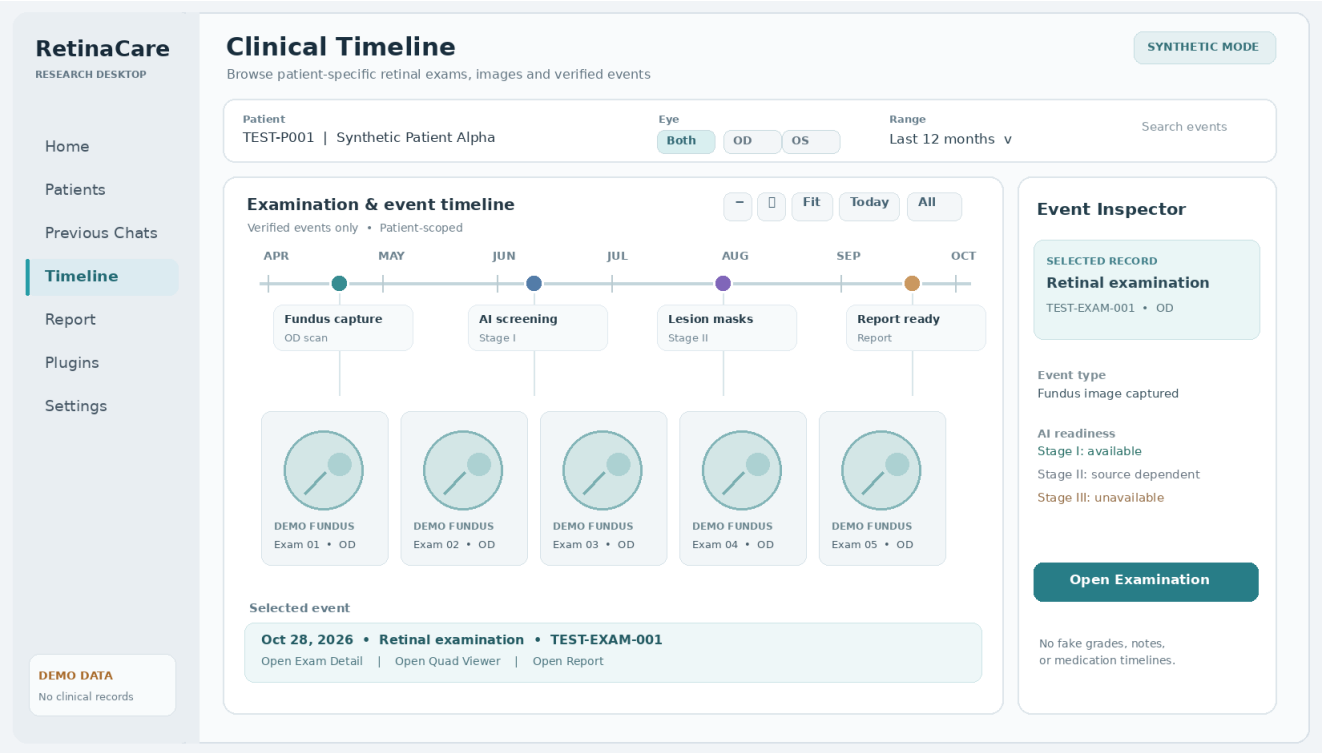
01 / Reference image - what to borrow



Your uploaded reference: a dense medical record timeline with left navigation, temporal lanes, scan thumbnails and a contextual right inspector. Treat it as layout inspiration, not a direct retinal product screenshot.

REFERENCE ELEMENT	RETINACARE ADAPTATION
Global left sidebar	Add Timeline as a distinct primary route, keeping Previous Chats separate.
Horizontal date ruler	Real exam/event timestamps, functional zoom, pan and fit to range.
Imaging strips	Authorized fundus thumbnails and mask availability, not brain MRI.
Medical notes panel	A truthful event inspector; no fabricated medicines, allergy lists or diagnosis text.
Dense black styling	Softer charcoal surfaces, restrained teal/blue accents, clearer visual hierarchy.

02 / Target layout - more breathable than reference

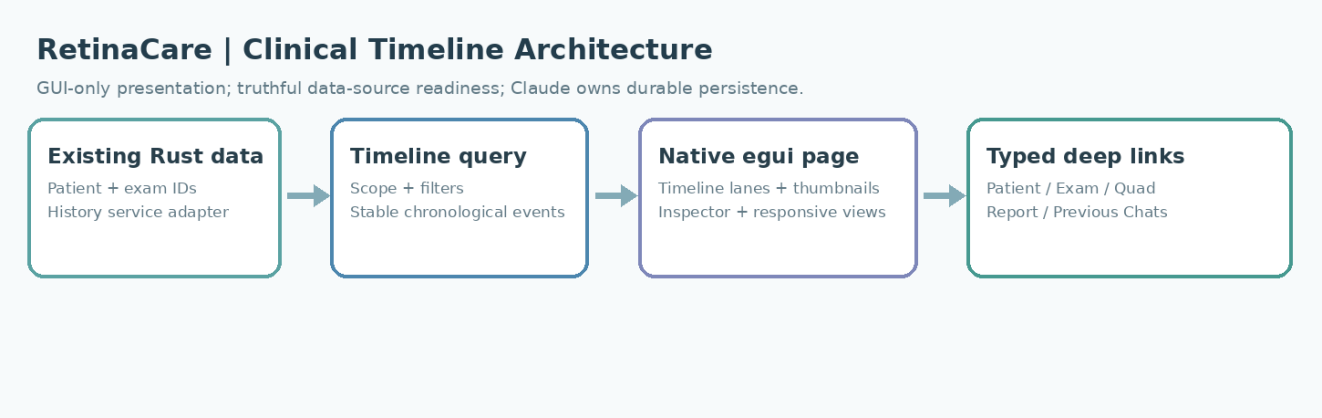


Suggested reading order: Sidebar > Patient and filter bar > Timeline ruler > Event lanes > Retinal thumbnail rail > Inspector and deep links. Controls must perform actual actions; unavailable sources must not be masked as empty history.

Design language

- Native egui/eframe only. Preserve global RetinaCare shell, typographic hierarchy and theme.
- Soft charcoal as existing theme allows; teal selection, subtle border, gentle rounding, 12-18px rhythm, no heavy gradients or glowing neon.
- Inspector around 260-320px on desktop; compact collapsible panel on medium screens; mobile-like single-column details on narrow windows.

03 / Data architecture and ownership



Antigravity owns UI and view-state mapping. Claude owns backend events, persistence, medical inference, audit and PDF services. No backend code edits, no shadow SQLite/JSON patient store and no new HTTP server.

04 / Two separate but connected history destinations

LEFT SIDEBAR ITEM	WHAT IT CONTAINS	HOW THEY CONNECT
Previous Chats	Conversations, safe previews, existing chat detail and optional persisted messages	Open linked examination in Timeline only when a real relationship exists
Timeline (NEW)	Chronological fundus exams, image events, screening and segmentation readiness	Open Exam Detail, Quad, Report, Patient and known conversations

Patient-scoped security

- Never combine records of different patients, even while asynchronous loads are in flight.
- Use stable patient_id + exam_id + event_id + eye + source references, not display names as keys.
- On patient switch invalidate stale results, thumbnails, inspector selection and queued callbacks.
- Open history must never rerun Stage I, Stage II, Stage III or export automatically.

05 / Feature specification

Patient-scope header

Selected patient, validated identity, source/fixture badge, eye/date filtering and transparent source readiness.

Timeline clock

Month/week/day tick spacing across range, zoom+/-, fit, scroll, same-date collision strategy, correct UTC/timezone.

Event tracks

Real exam / image / screening / mask / grading / report events only if backend provides them.

Thumbnail rail

Lazy loading of authorized retinal images, stable preview selection, missing-media fallback and direct Quad navigation.

Event inspector

Eye, exact timestamps, patient/exam IDs, source, truthful availability and functioning navigation actions.

History links

Previous Chats-to-Timeline and Timeline-to-Exam/Patient/Quad/Report using existing typed routes.

Responsive layout

Desktop at 1280x820, 1024x768 tablet, 900px boundary, 700x650 narrow and long-label tests.

Full UI states

Loading, source truly empty, no filter matches, unavailable, retryable error, stale selection and synthetic fixture.

Keyboard/idle

Tab/arrow/Enter/escape when native controls support them; real idle/sleep-resume verification separate from unit tests.

06 / Milestones - implemented only after plan

STEP	DELIVERABLE
T0	Inspect current code, Task 07 status, memory, routes, contracts; write 09-clinical-timeline-visual-plan.md.
T1	Create distinct primary sidebar Timeline route and stable patient/exam focus navigation.
T2	Implement source-backed filters, date ruler, zoom/fit and responsive event lanes.
T3	Wire authorized thumbnail rail, event selection/inspector and truthful readiness states.
T4	Integrate cross-navigation with Previous Chats/Patient/Exam/Quad/Report and strict privacy guards.
T5	Test all interactions, keyboard, patient transitions, many events, narrow/desktop and idle recovery.
T6	Run full Rust quality gates, launch actual binary, capture evidence, update GUI Obsidian and report.

07 / Required real tests

- Search/filter/eye scope/zoom/fit and event selection actually respond; no inert mock controls.
- All event dates and thumbnails come from real authorized records, or clearly labeled synthetic fixtures.
- Loading vs no history vs no matches vs unavailable vs error are visibly distinct.
- Same-day identical timestamps, mixed OD/OS, years of events, huge list, Unicode names, missing images.
- Patient A -> patient B transition ignores late A results, clears A image/inspector; no cross-patient leakage.
- Open event does not trigger inference, write records, fabricate grades or create unexpected audit data.
- Manual keyboard and real long-idle/resume tests logged honestly; 110 prior tests not assumed today.

08 / Verification commands and artifact proof

Confirm repository root first. At GUI workspace root, run:

```
cargo fmt --manifest-path rust/Cargo.toml --all -- --check
cargo check --manifest-path rust/Cargo.toml
cargo clippy --workspace --all-targets --manifest-path rust/Cargo.toml -- -D warnings
cargo test --workspace --manifest-path rust/Cargo.toml
cargo build --release --manifest-path rust/Cargo.toml
git diff --check
```

When working inside rust/, remove the rust/ prefix. Report executed commands and actual pass/fail/ignored results, not copied historical counts.

Required native screenshots

- timeline_desktop_verified.png - 1280x820 or wider, real running app
- timeline_tablet_verified.png - around 1024x768
- timeline_narrow_verified.png - around 700x650
- timeline_event_inspector_verified.png - event and thumbnail selection
- timeline_source_unavailable_verified.png - honest missing backend source
- timeline_navigation_verified.png - verified page/deep-link behavior when supported

Store documentation and images under docs/features/GUI-P07-previous-chats-history/. Update Antigravity GUI vault /home/claude/Desktop/Hackthon dezine/memory, keep notes concise with working wikilinks. No commit/push/merge without user permission.

09 / Critical non-goals and handoff

- Do not copy reference application names, branding, brain scans, medications, allergies, medical commentary or unspecified features.
- Do not redesign the whole app or convert it to web technologies; native Rust egui/eframe stays.
- Do not make Previous Chats disappear; Timeline is an additional separate destination.
- Do not implement a GUI-only clinical database; backend contracts belong to Claude.
- Do not claim functional PDF/audit/grading or saved conversations based on fixture display.
- Do not begin Task 08; Task 07.1 remains an addendum to existing Task 07.
- Do not commit, push or merge without permission.

Acceptance verdict rubric

VERDICT	MEANING
GUI complete	Routed page, functional available controls, responsive UI, privacy checks, tests and native screenshots pass.
Data integration partial	Real event/persistence/image source unavailable; GUI says so and supplies typed backend contract.
Blocked	Core GUI navigation, tests or source truthfulness are not functional.

10 / COMPLETE NORMAL TEXT MASTER PROMPT (copy-ready)

The entire normal prompt appears below and is also provided as a standalone UTF-8 TXT. Use this alongside the previous Task 07 guide and the attached visual reference.

RETINACARE - ANTIGRAVITY TASK 07.1 VISUAL ADDENDUM
CLINICAL TIMELINE / EXAMINATION HISTORY WORKSPACE (NATIVE RUST GUI)
READ TOGETHER WITH TASK 07 - PREVIOUS CHATS & HISTORY MASTER GUIDE

ROLE AND SCOPE
You are Antigravity, sole owner of the RetinaCare native Rust egui/eframe GUI. The user provides an additional screenshot of a dense medical-history timeline application. Use its INFORMATION ARCHITECTURE as inspiration: time ruler, event tracks, thumbnail strip, toolbar filters, contextual inspector, and layered history browsing. This is a visual reference, NOT an instruction to clone branding, MRI imagery, medical facts, or every control. Implement a softer, clearer, RetinaCare-specific clinical timeline without displacing the existing Task 07 Previous Chats work.

HIGHEST-PRIORITY USER CHANGE
Add a NEW, genuinely separate PRIMARY LEFT SIDEBAR NAVIGATION ITEM named "Timeline" (or "Clinical Timeline" if it fits). It must open a dedicated routed page inside the SAME native desktop window. Do not put the timeline inside Report, Patient Directory, a toolbar dropdown, a floating window, or the Previous Chats detail pane. Keep Previous Chats as its own separate Task 07 page. Both pages should interoperate through typed cross-navigation, not compete for the same route. Sidebar target order can be Home, Patients, Previous Chats, Timeline, Report, Plugins, Settings, adjusted only for existing conventions. Preserve every pre-existing page and feature.

DESIGN DIRECTION / HOW TO INTERPRET ATTACHED SCREENSHOT
The reference is a very dense dark medical imaging timeline. Borrow its layout logic: left global navigation; page-level filter bar; a real horizontal date ruler; dated milestones connected to examination event tracks; thumbnails linked to the selection; right-hand details / metadata inspector. Adapt to RETINAL FUNDUS images and authentic clinical-examination events only. Do NOT replicate brain MRI images, patient medications, allergies, clinical notes or diagnosis text from the example; do not present fabricated DR grades. Make the UI more comfortable and modern than the reference: refined native eframe styling, softer low-contrast surfaces, airy 12-18px spacing, calm teal/blue focus color, subtle dividers, 12-14px panel rounding where supported, clear visual hierarchy, large readable timestamp labels, restrained badges. Prefer a charcoal-soft mode matching RetinaCare's existing theme; a light mode must be opt-in/compatible with real app theme rather than a forced app-wide redesign. Do NOT redesign the global application shell, titlebar, font system, or other routes. The mockup in the supplement is conceptual, not a pixel-perfect demand.

READ FIRST; WRITE A PLAN BEFORE EDITING
1. Verify GUI repo /home/claude/Desktop/Hackthon dezine/rust and the actual Git HEAD/branch/worktree. Read GUI-owned Obsidian vault /home/claude/Desktop/Hackthon dezine/memory, start-here, active phase, Task 06.1 navigation handoff, and existing Task 07 implementation status. Check whether previous Task 07 prompt has already been executed; continue existing work instead of duplicating features.
2. Inspect typed AppPage navigation, patient/exam subroutes, Previous Chats route, data adapters and model/service contracts; inspect existing report/quad deep links and whether exam timestamps, original retinal thumbnails, OD/OS, event types and statuses are genuinely available.
3. Write docs/features/GUI-P07-previous-chats-history/09-clinical-timeline-visual-plan.md FIRST: route map, data-source readiness, UI wireframe, responsive states, source-backed event taxonomy, hooks to existing history, file-by-file modification plan, acceptance tests, and the backend interface gaps Claude must fill.
4. Implement after planning. Do not scan/rebuild the full app, create duplicate persistence, invent patient data, or accidentally start Task 08.

NEW DEDICATED TIMELINE PAGE - REQUIRED DESIGN
A. Top header: title Clinical Timeline, selected-patient scope with safe patient display identifier, status if synthetic/development mode, and a clear view of unavailable backend. Do not imply global cross-patient records are authorized.
B. Filter toolbar: patient selector if one already exists, laterality Both/OD/OS, source-backed event-type chips (Exam, Image Capture, Screening, Masks, Report, Note only when real), date range, search, sorting and Refresh only if truly supported. A disabled control must explain why. The screenshot reference is illustrative; do not show unusable decorative controls.
C. Central time ruler: chronological date/time axis, major month/week/day ticks according to span, meaningful zoom in/out/fit range controls, selected tick, pan/scroll or keyboard navigation. Items must be tied to actual known timestamps; never generate placeholder dates unless explicitly SYNTHETIC DEV FIXTURES. The timeline should stay readable when two events occur at the same time, event count is high, records span years, or no records exist. Patient identity scope always governs every query.
D. Event lanes: group records by individual examination and/or supported record type; clearly display Eye OD/OS, event identity, timestamp, status and source. Show truthful markers for fundus capture, Stage I screening, Stage II mask readiness, Stage III grading, PDF report only if the backend actually persisted those events. Absence of a model result must say Not available, never Grade 0.
E. Scan strip: load only real permitted retinal thumbnail assets linked to an examination, with lazy loading, caching and

missing-image fallbacks; do not use MRI images or copy the attached reference's visual content. Small previews open existing Exam Detail/Quad viewer, no rerunning models. Preserve original-image/mask provenance, source ID and laterality; overlay selection cannot silently cross patient scopes.
F. Right inspector: selected event summary with timestamp, patient/exam IDs, eye, source, model/service readiness and navigation buttons. Read-only by default. Only show observations, medications, notes, diagnoses, reports or attribution when authentic typed data exists and disclosure is permitted. Don't recreate the reference's free-text medication/clinical note sidebar from visual decoration.
G. Interactions: clicking an event highlights it, locates its thumbnail, updates the inspector and keeps stable IDs. Double-click/Enter invokes a real typed deep link. Clicking outside clears selection only if appropriate. Fit/zoom/toggle filters must respond to input and preserve valid selection. Back navigation returns to the prior page, patient and scroll position as feasible. Provide tooltip dates and empty states.
H. Responsive layout: at 1280x820, spacious 3-zone layout (sidebar + scrollable timeline and bottom thumbnail rail + ~260-320px inspector) with no giant dead zones; at 1024x768 shrink inspector or show as toggle; under 900px use a logical tab / full-width stacked list + selected event detail, not unreadable squashed timeline; at 700x650 and smaller no horizontal page overflow. Ensure timeline itself may scroll horizontally without dragging the global shell.
I. UX quality: native keyboard focus, tab/arrow/Enter/escape where existing widget support allows; actual hover tooltips, loss-of-focus recovery, mouse wheel semantics, sensible event-card hitboxes, accessibility contrast, text ellipsis with tooltips and long/unicode identifiers. No constantly blinking timelines, no excessive animations, no per-frame disk I/O and no high-frequency repaints. Preserve verified idle heartbeat/Glow configuration but manually test idle resume.

DATA AND SERVICE OWNERSHIP
Antigravity owns only GUI rendering/state/typed adapters; Claude owns Dird/native backend domain, persistence, medical inference, PDF and audit. The selected backend vault is /home/claude/Desktop/Hackthon dezine/Hackthon Backend. Do not alter or merge it. Reuse real patient/examination records and existing GUI client contracts. If durable exam events, image thumbnails, history listing, or provenance are not available, build the honest disabled/unavailable state and document explicit request/response structs for Claude to implement later. Do not create a shadow JSON or SQLite clinical database in the GUI, fake persistence, HTTP endpoints, or medical grades. Synthetic fixture mode requires a prominent SYNTHETIC/DEV indication. Use timestamps from the actual source and indicate timezone; do not silently shift UTC/local time.

INTEGRATION WITH EXISTING TASK 07
Previous Chats stays a separate sidebar destination. Implement typed links from a conversation's associated examination (when real) to Timeline focused on that exact exam, and from Timeline back to that conversation where a validated relationship exists. Patient Profile and Exam Detail can offer Open in Timeline; Timeline Inspector can offer Open Profile, Open Exam Detail, Open Quad, Open Report only when valid. Maintain patient/exam/eye scoping; never replay models merely by browsing or clicking timeline entries. Reuse existing report statuses rather than rebuilding Task 05 components. Do not introduce a new clinical note editor in Task 07.1.

STATE MODEL (ILLUSTRATIVE; ADAPT NAMES TO ACTUAL CODE)
TimelineRoute { patient_id: Option<PatientId>, focused_exam_id: Option<ExamId> }
TimelineViewState: Loading | Ready {events, selection} | Empty | NoMatches | Unavailable {reason} | Error {retryable}
TimelineEventSummary {stable_id, patient_id, exam_id, laterality, timestamp_utc, event_kind, source_state, thumbnail_ref?}
TimelineInteraction {time_range, zoom, selected_event_id, active_filters, inspector_open, scroll_offset}
Prevent stale async responses: patient or filter generation IDs must be checked before updating selection, images or inspector. Do not cache patient A details after switching to patient B. Build efficient sorted event index and visible-range filtering; avoid quadratic filtering and excessive per-frame allocations.

TEST PLAN / DEMONSTRATION
- Sidebar Timeline goes to a distinct page; Previous Chats, Patients, Report and Quad still work.
- Time scale: same-day events, identical timestamps, mixed OD/OS, multi-year records, zero events, large number of events, UTC/timezone boundaries and invalid dates.
- Search/filter/zoom/fit/inspect controls work (not decorative), selected-event thumb and inspector stay synchronized, correct patient+exam scoped navigation works.
- Selected patient A changes to B during async event or thumbnail load: patient A data must not render in B view.
- Empty vs NoMatches vs Unavailable vs Loading vs Error displayed honestly, backend-unavailable state nonempty and understandable.
- DEV fixture screenshots clearly label synthetic records. No invented clinical data or unlicensed imagery presented as actual patient history.
- Native responsive screenshots: timeline_desktop_verified.png at 1280x820; timeline_tablet_verified.png around 1024x768; timeline_narrow_verified.png around 700x650; timeline_no_data_verified.png; selected_event_inspector_verified.png; navigation_from_chat_verified.png only if link is supported. Screenshots must come from a running compiled native app, not a concept sketch.
- Verify existing History Task 07 privacy assertions, test no reruns when browsing; verify 20+ minutes idle and monitor wake/click if actually performed, else label unverified.
- Commands from verified GUI repo root: cargo fmt --manifest-path rust/Cargo.toml --all -- --check; cargo check --manifest-path rust/Cargo.toml; cargo clippy --workspace --all-targets --manifest-path rust/Cargo.toml -- -D warnings; cargo test --workspace --manifest-path rust/Cargo.toml; cargo build --release --manifest-path rust/Cargo.toml; git diff --check. If inside rust/ omit the prefix. Record exact commands/results and actual test counts; don't claim 110 is today's count without running tests.

IMPLEMENTATION MILESTONES

T0: Audit Task 07 progress, data sources, current navigation and build a written scoped plan.
T1: Add new primary sidebar Timeline route and stable patient/exam focused navigation.
T2: Implement softer timeline scaffold: header, real filters, scale, zoom, event lanes, responsive layout.
T3: Implement thumbnail rail, selected-event inspector and view-state handling, with honest source status.
T4: Cross-link Previous Chats, Patients, Exam Detail, Quad and Report where valid; enforce privacy/isolation.
T5: Real user interaction tests, keyboard navigation, large timelines, narrow windows, idle regression.
T6: Full Rust gate run, release build, actual desktop screenshots, docs, canonical GUI Obsidian handoff.
This is Task 07.1, a design/navigation ADDENDUM to Task 07, not Task 08 and not a new backend implementation phase.

FINAL REPORT AND ACCEPTANCE

Report branch/HEAD + clean/dirty status; exact files changed; new sidebar route and screenshot proof; checklist of working controls; source authenticity and backend contracts; all tests including ignored; privacy proof; responsive screenshots; report/quad regression; Obsidian notes and wikilinks; and COMPLETE/PARTIAL/BLOCKED split between GUI work and actual durable history data. Do not commit/push without direct user authorization. Do not modify Claude backend or claim clinical/diagnostic validation. Do not stop at a design-only explanation: implement the native GUI page and verify it.

ACCEPTANCE CHECKLIST

[ ] Dedicated Timeline item in primary LEFT SIDEBAR is clickable and opens its own routed page.
[ ] Previous Chats remains a separate working Task 07 page; no routes overwritten.
[ ] Softer premium RetinaCare look inspired by layout, not by MRI or copied brand styles.
[ ] Time ruler, event lanes, selection, inspector and thumbnails appear only from authorized source-backed records or clearly marked synthetic fixtures.
[ ] Search, OD/OS filter, source-backed event types, time range, zoom/fit and detail navigation WORK.
[ ] Responsive 1280x820, 1024x768 and 700x650 views have no unreadable compression or dead zones.
[ ] Correct loading, empty, no matches, unavailable, error and stale event states.
[ ] Cross-patient scope isolation and no automatic AI reruns on timeline navigation.
[ ] No fake grades, medical notes, allergies, medications or reports; no new GUI-owned clinical persistence.
[ ] Existing sidebar, quad/report/patients/chat behaviors and idle fix preserved.
[ ] Run fmt/check/strict clippy/test/release/diff, provide exact outputs and native screenshots.
[ ] Canonical Antigravity Obsidian notes, roadmap and handoff updated; source links resolve.
[ ] Honest outcome: visual Task 07.1 can be done while backend history remains unavailable.
[ ] No commits/push/merge without authorization; do not begin Task 08.

11 / COMPLETE JSON MASTER PROMPT

The complete structured JSON is included below and delivered separately for direct agent use. The JSON full_normal_prompt repeats the normal prompt for self-contained handoff.


```
{
"title": "RetinaCare Antigravity Task 07.1 Clinical Timeline Visual Addendum",
"read_with": "RetinaCare Antigravity Task 07 Previous Chats & History Master Guide",
"source_reference": {
"uploaded_image_filename": "image(20261028-055609).png",
"purpose": "Information architecture reference only: timeline + date ruler + thumbnails + inspector",
"do_not_copy": [
"branding",
"MRI imagery",
"clinical note text",
"allergy or medication claims",
"unrelated controls"
]
},
"owner": "Antigravity (native Rust egui/eframe GUI only)",
"backend_owner": "Claude; do not edit Dird/native",
"goal": "Add a new primary LEFT SIDEBAR Timeline page, separate from Previous Chats, with source-backed exam timeline and soft premium RetinaCare-specific UI.",
"paths": {
"gui_repo_expected": "/home/claude/Desktop/Hackthon dezine/rust",
"gui_vault": "/home/claude/Desktop/Hackthon dezine/memory",
"backend_repo_read_only": "/home/claude/Desktop/Dird"
},
"required_features": [
"new typed Timeline sidebar route",
"time axis with functional zoom/fit/pan",
"scoped event lanes per patient/exam/OD-OS",
"real timestamp handling",
"source-backed filters and search",
"retinal image thumbnail strip with honest unavailable fallback",
"selected event inspector",
"responsive desktop/tablet/narrow layout",
"cross-link to Chats/Patient/Exam/Quad/Report",
"loading/empty/no-matches/unavailable/error distinctions",
"no reruns or cross-patient leaks",
"native keyboard interaction and accessibility r
eview"
],
"design": {
"theme": "existing soft charcoal native egui UI; airy surfaces, restrained teal accent, 12-14px panel radii if supported",
"desktop": "sidebar + timeline + thumbnail rail + 260-320px inspector",
"tablet": "resizable/toggle inspector",
"narrow": "single column list and detail view; no horizontal global overflow"
},
"implementation_milestones": [
{
"id": "T0",
"goal": "Read memory/source and write plan"
},
{
"id": "T1",
"goal": "New sidebar route + typed navigation"
},
{
"id": "T2",
"goal": "Timeline ruler/filters/lanes and responsive shell"
},
{
"id": "T3",
```

"goal": "Thumbnail rail + inspector + truthful view states"
},
{
"id": "T4",
"goal": "Cross-route links + privacy/epoch guards"
},
{
"id": "T5",
"goal": "UI/keyboard/responsive/idle regression tests"
},
{
"id": "T6",
"goal": "fmt/check/clippy/test/release/screenshots/docs/vault handoff"
}
],
"tests": [
"sidebar route does not replace Previous Chats",
"same timestamp, mixed eye, large dataset, timezone, no data",
"functional zoom/filter/search/inspect",
"stale asynchronous patient scope isolation",
"unavailable state distinct from zero records",
"real screenshot proof at 1280x820 1024x768 700x650",
"no model reruns when browsing",
"test gates with exact results",
"Obsidian links resolve"
],
"never": [
"edit Claude backend",
"fabricate saved data or medical results",
"use MRI scan as Re

tinaCare result",
"commit or push without approval",
"start Task 08",
"replace existing application visual style",
"invent a second desktop window"
],
"full_normal_prompt": "RETINACARE - ANTIGRAVITY TASK 07.1 VISUAL ADDENDUM\nCLINICAL TIMELINE / EXAMINATION HISTORY WORKSPACE (NATIVE RUST GUI)\nREAD TOGETHER WITH TASK 07 - PREVIOUS CHATS & HISTORY MASTER GUIDE\n\nROLE AND SCOPE\nYou are Antigravity, sole owner of the RetinaCare native Rust egui/eframe GUI. The user provides an additional screenshot of a dense medical-history timeline application. Use its INFORMATION ARCHITECTURE as inspiration: time ruler, event tracks, thumbnail strip, toolbar filters, contextual inspector, and layered history browsing. This is a visual reference, NOT an instruction to clone branding, MRI imagery, medical facts, or every control. Implement a softer, clearer, RetinaCare-specific clinical timeline without displacing the existing Task 07 Previous Chats work.\n\nHIGHEST-PRIORITY USER CHANGE\nAdd a NEW, genuinely separate PRIMARY LEFT SIDEBAR NAVIGATION ITEM named \"Timeline\" (or \"Clinical Timeline\" if it fits). It must open a dedicated routed page inside the SAME native desktop window. Do not put the timeline inside Report, Patient Directory, a toolbar dropdown, a floating window, or the Previous Chats detail pane. Keep Previous Chats as its own separate Task 07 page. Both pages should interoperate through typed cross-navigation, not compete for the same route. Sidebar target order can be Home, Patients, Previous Chats, Timeline, Report, Plugins, Settings, adjusted only f

or existing conventions. Preserve every pre-existing page and feature.\n\nDESIGN DIRECTION / HOW TO INTERPRET ATTACHED SCREENSHOT\nThe reference is a very dense dark medical imaging timeline. Borrow its layout logic: left global navigation; page-level filter bar; a real horizontal date ruler; dated milestones connected to examination event tracks; thumbnails linked to the selection; right-hand details / metadata inspector. Adapt to RETINAL FUNDUS images and authentic clinical-examination events only. Do NOT replicate brain MRI images, patient medications, allergies, clinical notes or diagnosis text from the example; do not present fabricated DR grades. Make the UI more comfortable and modern than the reference: refined native eframe styling, softer low-contrast surfaces, airy 12-18px spacing, calm teal/blue focus color, subtle dividers, 12-14px panel rounding where supported, clear visual hierarchy, large readable timestamp labels, restrained badges. Prefer a charcoal-soft mode matching RetinaCare's existing theme; a light mode must be opt-in/compatible with real app theme rather than a forced app-wide redesign. Do NOT redesign the global application shell, titlebar, font system, or other routes. The mockup in the supplement is conceptual, not a pixel-perfect demand.\n\nREAD FIRST; WRITE A PLAN BEFORE EDITING\n1. Verify GUI repo /home/claude/Desktop/Hackthon dezine/rust and the actual Git HEAD/branch/worktree. Read GUI-owned Obsidian vault /home/claude/Desktop/Hackthon dezine/memory, start-here, active phase, Task 06.1 navigation handoff, and existing Task 07 implementation

status. Check whether previous Task 07 prompt has already been executed; continue existing work instead of duplicating features.\n2. Inspect typed AppPage navigation, patient/exam subroutes, Previous Chats route, data adapters and model/service contracts; inspect existing report/quad deep links and whether exam timestamps, original retinal thumbnails, OD/OS, event types and statuses are genuinely available.\n3. Write docs/features/GUI-P07-previous-chats-history/09-clinical-timeline-visual-plan.md FIRST: route map, data-source readiness, UI wireframe, responsive states, source-backed event taxonomy, hooks to existing history, file-by-file modification plan, acceptance tests, and the backend interface gaps Claude must fill.\n4. Implement after planning. Do not scan/rebuild the full app, create duplicate persistence, invent patient data, or accidentally start Task 08.\n\nNEW DEDICATED TIMELINE PAGE - REQUIRED DESIGN\nA. Top header: title Clinical Timeline, selected-patient scope with safe patient display identifier, status if synthetic/development mode, and a clear view of unavailable backend. Do not imply global cross-patient records are authorized.\nB. Filter toolbar: patient selector if one already exists, laterality Both/OD/OS, source-backed event-type chips (Exam, Image Capture, Screening, Masks, Report, Note only when real), date range, search, sorting and Refresh only if truly supported. A disabled control must explain why. The screenshot reference is illustrative; do not show unusable decorative controls.\nC. Central time ruler: chronological date/time axis, major mon

th/week/day ticks according to span, meaningful zoom in/out/fit range controls, selected tick, pan/scroll or keyboard navigation. Items must be tied to actual known timestamps; never generate placeholder dates unless explicitly SYNTHETIC DEV FIXTURES. The timeline should stay readable when two events occur at the same time, event count is high, records span years, or no records exist. Patient identity scope always governs every query.\nD. Event lanes: group records by individual examination and/or supported record type; clearly display Eye OD/OS, event identity, timestamp, status and source. Show truthful markers for fundus capture, Stage I screening, Stage II mask readiness, Stage III grading, PDF report only if the backend actually persisted those events. Absence of a model result must say Not available, never Grade 0.\nE. Scan strip: load only real permitted retinal thumbnail assets linked to an examination, with lazy loading, caching and missing-image fallbacks; do not use MRI images or copy the attached reference's visual content. Small previews open existing Exam Detail/Quad viewer, no rerunning models. Preserve original-image/mask provenance, source ID and laterality; overlay selection cannot silently cross patient scopes.\nF. Right inspector: selected event summary with timestamp, patient/exam IDs, eye, source, model/service readiness and navigation buttons. Read-only by default. Only show observations, medications, notes, diagnoses, reports or attribution when authentic typed data exists and disclosure is permitted. Don't recreate the reference's free-text medicati

on/clinical note sidebar from visual decoration.\nG. Interactions: clicking an event highlights it, locates its thumbnail, updates the inspector and keeps stable IDs. Double-click/Enter invokes a real typed deep link. Clicking outside clears selection only if appropriate. Fit/zoom/toggle filters must respond to input and preserve valid selection. Back navigation returns to the prior page, patient and scroll position as feasible. Provide tooltip dates and empty states.\nH. Responsive layout: at 1280x820, spacious 3-zone layout (sidebar + scrollable timeline and bottom thumbnail rail + ~260-320px inspector) with no giant dead zones; at 1024x768 shrink inspector or show as toggle; under 900px use a logical tab / full-width stacked list + selected event detail, not unreadable squashed timeline; at 700x650 and smaller no horizontal page overflow. Ensure timeline itself may scroll horizontally without dragging the global shell.\nI. UX quality: native keyboard focus, tab/arrow/Enter/escape where existing widget support allows; actual hover tooltips, loss-of-focus recovery, mouse wheel semantics, sensible event-card hitboxes, accessibility contrast, text ellipsis with tooltips and long/unicode identifiers. No constantly blinking timelines, no excessive animations, no per-frame disk I/O and no high-frequency repaints. Preserve verified idle heartbeat/Glow configuration but manually test idle resume.\n\nDATA AND SERVICE OWNERSHIP\nAntigravity owns only GUI rendering/state/typed adapters; Claude owns Dird/native backend domain, persistence, medical inference, PDF and audit. The select

ed backend vault is /home/claude/Desktop/Hackthon dezine/Hackthon Backend. Do not alter or merge it. Reuse real patient/examination records and existing GUI client contracts. If durable exam events, image thumbnails, history listing, or provenance are not available, build the honest disabled/unavailable state and document explicit request/response structs for Claude to implement later. Do not create a shadow JSON or SQLite clinical database in the GUI, fake persistence, HTTP endpoints, or medical grades. Synthetic fixture mode requires a prominent SYNTHETIC/DEV indication. Use timestamps from the actual source and indicate timezone; do not silently shift UTC/local time.\n\nINTEGRATION WITH EXISTING TASK 07\nPrevious Chats stays a separate sidebar destination. Implement typed links from a conversation's associated examination (when real) to Timeline focused on that exact exam, and from Timeline back to that conversation where a validated relationship exists. Patient Profile and Exam Detail can offer Open in Timeline; Timeline Inspector can offer Open Profile, Open Exam Detail, Open Quad, Open Report only when valid. Maintain patient/exam/eye scoping; never replay models merely by browsing or clicking timeline entries. Reuse existing report statuses rather than rebuilding Task 05 components. Do not introduce a new clinical note editor in Task 07.1.\n\nSTATE MODEL (ILLUSTRATIVE; ADAPT NAMES TO ACTUAL CODE)\nTimelineRoute { patient_id: Option<PatientId>, focused_exam_id: Option<ExamId> }\nTimelineViewState: Loading | Ready {events, selection} | Empty | NoMatches | Unavailable {

reason} | Error {retryable}\nTimelineEventSummary {stable_id, patient_id, exam_id, laterality, timestamp_utc, event_kind, source_state, thumbnail_ref?}\nTimelineInteraction {time_range, zoom, selected_event_id, active_filters, inspector_open, scroll_offset}\nPrevent stale async responses: patient or filter generation IDs must be checked before updating selection, images or inspector. Do not cache patient A details after switching to patient B. Build efficient sorted event index and visible-range filtering; avoid quadratic filtering and excessive per-frame allocations.\n\nTEST PLAN / DEMONSTRATION\n- Sidebar Timeline goes to a distinct page; Previous Chats, Patients, Report and Quad still work.\n- Time scale: same-day events, identical timestamps, mixed OD/OS, multi-year records, zero events, large number of events, UTC/timezone boundaries and invalid dates.\n- Search/filter/zoom/fit/inspect controls work (not decorative), selected-event thumb and inspector stay

synchronized, correct patient+exam scoped navigation works.\n- Selected patient A changes to B during async event or thumbnail load: patient A data must not render in B view.\n- Empty vs NoMatches vs Unavailable vs Loading vs Error displayed honestly, backend-unavailable state nonempty and understandable.\n- DEV fixture screenshots clearly label synthetic records. No invented clinical data or unlicensed imagery presented as actual patient history.\n- Native responsive screenshots: timeline_desktop_verified.png at 1280x820; timeline_tablet_verified.png around 1024x768; timeline_narrow_verified.png around 700x650; time

line_no_data_verified.png; selected_event_inspector_verified.png; navigation_from_chat_verified.png only if link is supported. Screenshots must come from a running compiled native app, not a concept sketch.\n- Verify existing History Task 07 privacy assertions, test no reruns when browsing; verify 20+ minutes idle and monitor wake/click if actually performed, else label unverified.\n- Commands from verified GUI repo root: cargo fmt --manifest-path rust/Cargo.toml --all -- --check; cargo check --manifest-path rust/Cargo.toml; cargo clippy --workspace --all-targets --manifest-path rust/Cargo.toml -- -D warnings; cargo test --workspace --manifest-path rust/Cargo.toml; cargo build --release --manifest-path rust/Cargo.toml; git diff --check. If inside rust/ omit the prefix. Record exact commands/results and actual test counts; don't claim 110 is today's count without running tests.\n\nIMPLEMENTATION MILESTONES\nT0: Audit Task 07 progress, data sources, current navigation and build a written scoped plan.\nT1: Add new primary sidebar Timeline route and stable patient/exam focused navigation.\nT2: Implement softer timeline scaffold: header, real filters, scale, zoom, event lanes, responsive layout.\nT3: Implement thumbnail rail, selected-event inspector and view-state handling, with honest source status.\nT4: Cross-link Previous Chats, Patients, Exam Detail, Quad and Report where valid; enforce privacy/isolation.\nT5: Real user interaction tests, keyboard navigation, large timelines, narrow windows, idle regression.\nT6: Full Rust gate run, release build, actual desktop screenshots

, docs, canonical GUI Obsidian handoff.\nThis is Task 07.1, a design/navigation ADDENDUM to Task 07, not Task 08 and not a new backend implementation phase.\n\nFINAL REPORT AND ACCEPTANCE\nReport branch/HEAD + clean/dirty status; exact files changed; new sidebar route and screenshot proof; checklist of working controls; source authenticity and backend contracts; all tests including ignored; privacy proof; responsive screenshots; report/quad regression; Obsidian notes and wikilinks; and COMPLETE/PARTIAL/BLOCKED split between GUI work and actual durable history data. Do not commit/push without direct user authorization. Do not modify Claude backend or claim clinical/diagnostic validation. Do not stop at a design-only explanation: implement the native GUI page and verify it.\n\nACCEPTANCE CHECKLIST\n[ ] Dedicated Timeline item in primary LEFT SIDEBAR is clickable and opens its own routed page.\n[ ] Previous Chats remains a separate working Task 07 page; no routes overwritten.\n[ ] Softer premium RetinaCare look inspired by layout, not by MRI or copied brand styles.\n[ ] Time ruler, event lanes, selection, inspector and thumbnails appear only from authorized source-backed records or clearly marked synthetic fixtures.\n[ ] Search, OD/OS filter, source-backed event types, time range, zoom/fit and detail navigation WORK.\n[ ] Responsive 1280x820, 1024x768 and 700x650 views have no unreadable compression or dead zones.\n[ ] Correct loading, empty, no matches, unavailable, error and stale event states.\n[ ] Cross-patient scope isolation and no automatic AI reruns on timeline navigat

ion.\n[ ] No fake grades, medical notes, allergies, medications or reports; no new GUI-owned clinical persistence.\n[ ] Existing sidebar, quad/report/patients/chat behaviors and idle fix preserved.\n[ ] Run fmt/check/strict clippy/test/release/diff, provide exact outputs and native screenshots.\n[ ] Canonical Antigravity Obsidian notes, roadmap and handoff updated; source links resolve.\n[ ] Honest outcome: visual Task 07.1 can be done while backend history remains unavailable.\n[ ] No commits/push/merge without authorization; do not begin Task 08.\n"
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